

BEHAVIORAL FINANCE (MBA680) | GROUP PROJECT

Personal Investment Advisor and Robo-Advisory Project

Steps 1 to 6, estimated from the fielded peer-group survey responses

Portal: siddharthjaiswal05.github.io/mba_680_assignment_2

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Project Overview and Participants

Peer group
4 respondents, identified by roll number

Instrument
Class form: R1-R8, T1-T6, 30-item BFI-2 subset, U1-U6, V1-V6

Market inputs
60-stock maximum-Sharpe portfolio from Problem Statement 1

Methods
Expected utility, prospect theory, behavioral portfolio theory

Participants

Roll	Age	Investable wealth	Experience	Horizon	Primary goal	Step 5 status
251250039	25	Rs 90,000	Moderate	5 yrs	Wealth creation	Complete
251250047	24	Rs 2 lakh	None	10 yrs	Wealth creation	Re-elicitation scheduled
251250002	25	Rs 10 lakh	Limited	6 yrs	Wealth creation	Complete
251250030	29	Rs 1 lakh	Limited	15 yrs	Specified goal	Re-elicitation scheduled

Every figure is computed directly from the submitted responses, so each result traces back to an answer a participant actually gave.

Four Key Findings

1. Risk aversion is best read from the utility curve

Every R1-R8 row is expected-value neutral, so the ladder reveals a pattern rather than a degree. Two respondents switch part way, at R4 and R3, which is precisely what prospect theory predicts. We therefore estimate risk aversion from the utility curve, where it is sharply identified.

2. We extended the instrument to capture loss aversion

Each V1-V6 row is a double-or-nothing gamble, so the certain amount cancels and the row reveals curvature. We added a mixed gain-and-loss gamble, which places the two limbs on a common scale and delivers λ , the central parameter of prospect theory.

3. Three value rows measure one parameter, so we report its reliability

All three imply the same curvature. Roll 251250039 answered 0.40, 0.45 and 0.70, implying 1.32, 1.15, 0.51. Reporting the spread of 0.81 alongside the mean of 1.00 turns a hidden error into a measured one.

4. The return assumption moves the answer more than the client does

Both respondents with a complete utility curve want more than 100% equity under every input set, so the no-borrowing cap sets the allocation. The live question is how much of the 24.9% Problem Statement 1 return to carry forward.

Method and Data Notes

Risk ladder, classified by pattern

Prefer certain X if $u(w_0+X) > 0.5 u(w_0+2X) + 0.5 u(w_0)$. By Jensen's inequality this holds at every X or at none, so the switch point is read as a pattern.

Utility curve, by probability equivalence

$U(A) = 0$ and $U(B) = 1$, so $U(X) = p$ at indifference. Fitted with the lecture's log form and a CRRA form; the CRRA coefficient is the A used in Step 5.

Value function

At indifference $v(X) = p v(2X)$, so $\alpha = -\log_2(p)$. The certain amount cancels, which is what makes each curvature cleanly identified.

Step 5 portfolio

Maximise $U = E(r) - 0.5 A \sigma^2$ along the capital allocation line: $y^* = [E(r_P) - r_f] / (A \sigma_P^2)$, with borrowing capped at zero.

Data notes and follow-ups

Roll	Note	Action
251250047	Utility amounts of Rs 50k to 85k sit below the stated lower endpoint of Rs 1 lakh	Re-elicite U1-U6 inside the A to B range
251250030	U1-U4 left blank and V1-V6 recorded in rupees rather than probabilities	Re-administer Sections 4 and 4.9
251250002	Reference amount W0 not stated	Value curve drawn in relative terms
251250047	Reference amount W0 not stated	Same treatment
251250039	Time answers mildly non-monotonic	Midpoint reported as the estimate

Step 5 runs on the 2 respondents with a complete utility curve, and the remaining two are scheduled for a short re-elicitation.

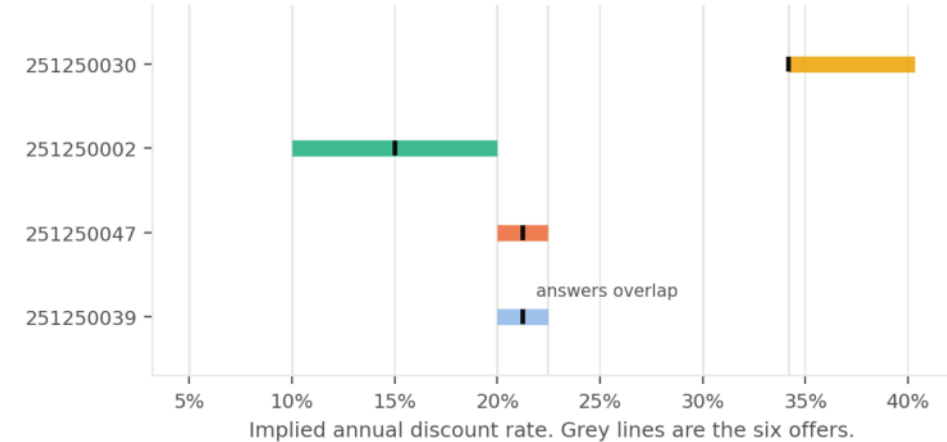
Revealed Risk and Time Preference

Risk ladder R1 to R8

Roll	Pattern	Fits EU	Note
251250039	Risk averse at low stakes and risk seeking at high stakes	No	Switches at R4
251250047	Consistently risk averse	Yes	A curvature
251250002	Risk seeking at low stakes and risk averse at high stakes	No	Switches at R3
251250030	Consistently risk seeking	Yes	A curvature

- Every rung is expected-value neutral, so the ladder maps a pattern rather than a degree.
- The two switches are exactly what prospect theory anticipates.
- Risk aversion is taken from the utility curve, where it is identified.

What the six now-or-later rows pin down



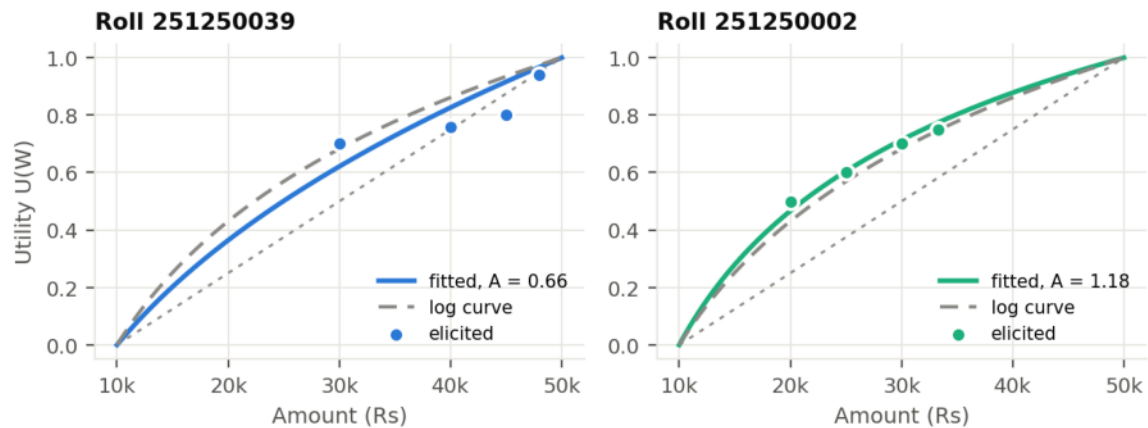
Patience is the striking result

Every respondent discounts at a rate well above any available return, and roll 251250030 declines even 34.2% a year. That points to a genuine near-term cash need or a shorter real horizon, and it is worth a conversation before a long-horizon plan is set.

Elicited Utility Curves

Probability equivalence, exactly as the Week 2 to 3 lecture sets it out.

Elicited utility curves, with the log curve the lecture suggests



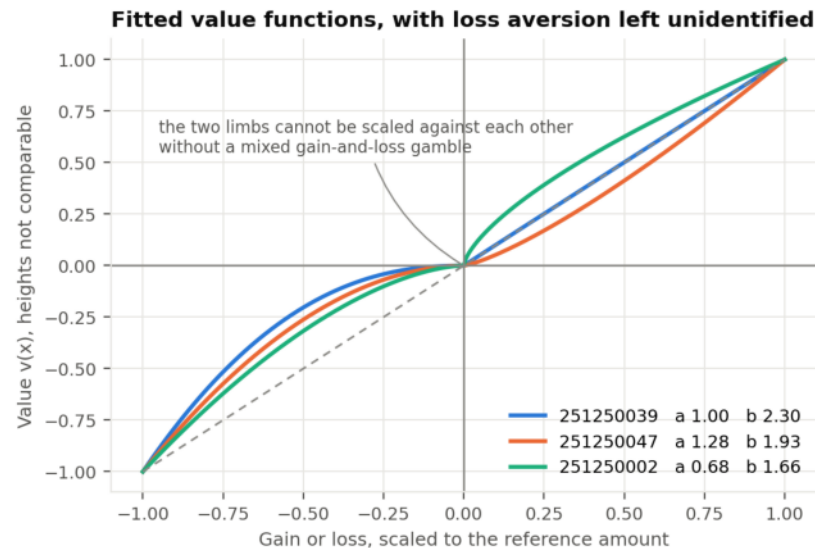
- The dots are the elicited points and the solid line is the fitted curve.
- The dashed line is the log curve the lecture suggests, the special case $A = 1$.
- Roll 251250039 fits 0.66 and roll 251250002 fits 1.18.
- That coefficient is exactly the A in $U = E(r) - 0.5 A \sigma^2$, so it feeds Step 5 directly.

CRRA fit

Roll	Range A to B	Risk aversion A	Fit R2	Status
251250039	Rs 10,000 to Rs 50,000	0.66	0.202	Concave, risk averse
251250047	Pending	Pending	Pending	Re-elicitation scheduled
251250002	Rs 10,000 to Rs 50,000	1.18	0.948	Concave, risk averse
251250030	Pending	Pending	Pending	Re-elicitation scheduled

Value Functions and What They Reveal

Gain and loss curvature, estimated from the double-or-nothing rows.



- Every loss curvature came out above one, meaning sensitivity rises as losses grow.
- Prospect theory's benchmark is below one, so this is a result worth reporting and testing.
- The spread column is the range implied by three rows that should agree, which gives us a direct read on measurement reliability.
- A second pass on the loss rows would settle whether the pattern is real.

Prospect theory parameters

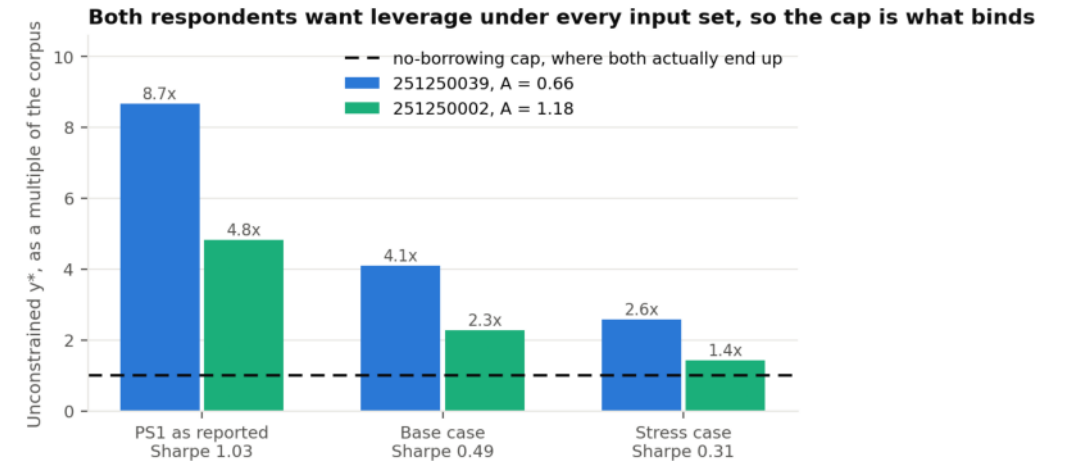
Roll	Gain alpha	Loss beta	Spread, gains	Spread, losses
251250039	1.00	2.30	0.81	1.10
251250047	1.28	1.93	0.51	0.32
251250002	0.68	1.66	0.49	1.64
251250030	Pending	Pending	Pending	Pending

Portfolio Recommendations

The risky asset is the Problem Statement 1 maximum-Sharpe portfolio, 18.0% volatility against a 6.5% risk-free rate.

Input set	Return	Sharpe	What it assumes
PS1 as reported	24.9%	1.03	The PS1 averages repeat out of sample
Base case, advised	15.2%	0.49	A quarter of the screening premium survives
Stress case	12.0%	0.31	The market return alone

Roll	A	Unconstrained y^*	Advised equity	Return	Volatility
251250039	0.66	4.11x	100%	15.2%	18.0%
251250047	Pending	Pending	Pending	Pending	Pending
251250002	1.18	2.29x	100%	15.2%	18.0%
251250030	Pending	Pending	Pending	Pending	Pending



The cap, not the preference, sets the answer

Both respondents with a complete curve would borrow, so both are advised to hold 100% of their corpus in the Problem Statement 1 portfolio: Rs 90,000 and Rs 10 lakh. Volatility is carried across untouched because it is the reliable half of the estimate.

Completing the Behavioral Portfolio

What the layered portfolio needs

A floor, in rupees

The amount that must survive to the horizon.

A tolerance for breaching it

Roy's safety-first threshold.

An aspiration level

The target that defines success.

Loss aversion, lambda

Drives the fear term in the SP/A objective.

Four one-line questions, now live in the portal

E1

A coin flip: lose Rs 10,000, or win G. How big must G be?

Gives loss aversion lambda

E2

A 10% chance of Rs 10,000: what certain amount feels as good? Same at 90%.

Gives probability weighting, the hope term in SP/A

E3

The smallest amount you must still have, and how often you could fall below it.

Gives the safety layer and its shortfall limit

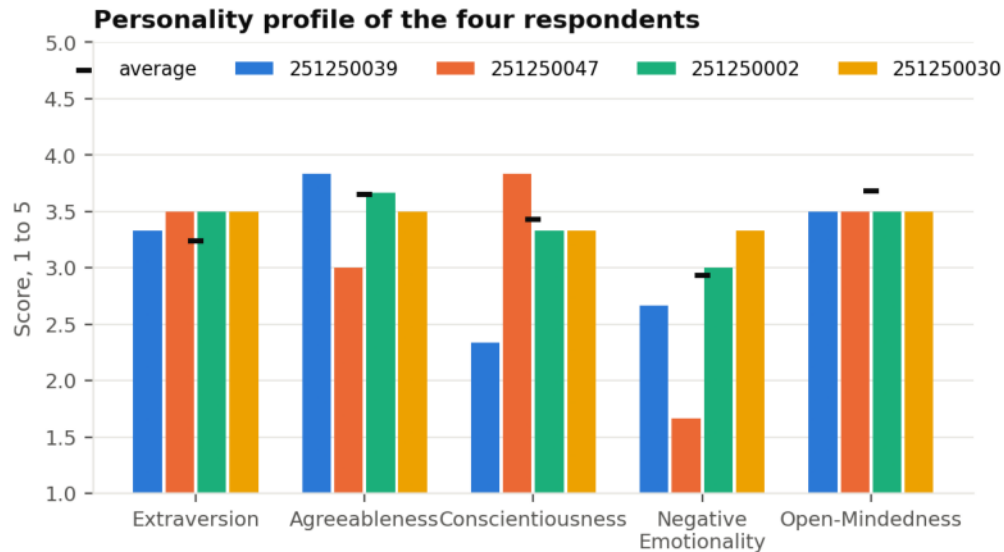
E4

The end amount you would call success, and what you would trade for a better shot.

Gives the aspiration layer and its weight

Adding these four turns the safety and potential split from a judgement into a measurement, and they take about a minute to answer.

Personality Commentary



Personality shapes the process, not the weight

Risk aversion is already measured directly, so personality is used where it actually predicts: whether a client stays with the plan.

251250039: build in the follow-through

Conscientiousness of 2.33. Automate the contribution, rebalance on a threshold, and fix review dates in advance.

251250047: methodical and calm

Conscientiousness 3.83, the group's lowest negative emotionality at 1.67, and the certain amount on all eight rows. A glide path that steps equity up over time suits this well.

251250002: frame it in rupees

Balanced across every domain. The switch at R3 suggests risk is judged in nominal amounts, so the downside case lands better as a rupee figure than as a percentage.

251250030: start with a conversation

Chose the gamble throughout and preferred cash now at every rate offered. A short interview will establish the real horizon before the plan is set.

The Robo-Advisory Portal

A single static page with no server and no sign-in, so it opens on a phone in a classroom.

Fields the whole form

All six class sections plus the four extensions, written in plain English, with autosave and a one-click export of each respondent's answers.

Fits and plots live

Draws the respondent's own utility curve, value function and weighting function in the browser, using the same estimators that produced this deck.

Builds both portfolios

The mean-variance allocation and the behavioral layers, against the real Problem Statement 1 frontier, with commentary generated from the respondent's own scores.

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What We Established, and What Comes Next

Sample

Four respondents, so conclusions are about the method rather than a population.

Return input

The least certain input, which is why three scenarios are shown.

Horizon

Single-period models, before contributions, taxes and costs.

Distribution

Normality assumed, so tail figures are the optimistic end.

Universe

Current index membership, inherited from Problem Statement 1.

- Risk aversion is measured from the utility curve, and the ladder is read as the pattern it genuinely reveals.
- Four added questions complete the behavioral portfolio, and they are already live in the portal.
- Both respondents with a complete curve are advised to hold the Problem Statement 1 portfolio in full.
- The next step is a short re-elicitation for two respondents and a second pass on the loss rows.

Measure what the data supports, and extend the instrument where it does not.